Assignment 2: Detecting Emotions with ML Algorithms

Cross-Dataset Generalization of Facial Expressions

Su Acar (2475901)
Vikram Chandrasekaran (2542544)
Henrique DaSilva Couto (2463334)

5ARE0 – Data Analysis & Learning Methods
Assignment 2 – Group 5
October 25, 2026

Example citations for later on (remove this later): (Goodfellow et al., 2013)
(Goeleven et al., 2008)
(Goeleven et al., 2008; Goodfellow et al., 2013)

1 Introduction and research questions

This report studies facial expression recognition with classical machine learning on two datasets, FER2013 and KDEF, and asks which challenges are specific to each dataset and how well knowledge transfers between them. The FER2013 baseline classifiers are covered in Task 1; Tasks 2 to 6 cover the KDEF study, cross-dataset generalization, the fuzzy classifier, the creative challenge and the real-time demo. *[Replace: One or two sentences summarising the overall main findings once all tasks are done.]*

The primary question is how datasets and feature/model choices affect within-dataset accuracy and cross-dataset generalization. Secondary questions concern which emotions and representations are most transferable, whether interpretable facial geometry is useful, and how reliable a trained model remains under webcam conditions. *[Replace: Add source-supported motivation and APA citations to FER2013/KDEF after Zotero export.]*

2 Datasets, preprocessing and experimental protocol

FER2013. Images are grayscale, 48×48, with seven labels: angry, disgust, fear, happy, sad, surprise and neutral. Use the provided Training split to fit models, PublicTest for model selection/tuning, and untouched PrivateTest *only* for final evaluation.

KDEF. Use only straight/frontal (S) faces, across seven comparable expressions (KDEF "Afraid" maps to FER "fear"). Divide by identity: 80% of subjects for training, 20% for testing, with no subject overlap.

Common representation. Detect and crop KDEF faces with a margin preserving forehead and jaw; convert to grayscale, resize to 48×48 and apply the same normalization as FER. Apply identical feature extraction at training and inference. Fit any scaling, PCA or learned transformation only on training data.

HOG features. Histogram of Oriented Gradients (HOG) looks at the edges in an image and identifies where the brightness changes and in which direction. To extract the features, the 48×48 grayscale image is cut into a 6×6 grid of cells of 8×8 pixels. In each cell, the change in brightness is measured in 9 different directions $(0°, 20°, \ldots, 160°)$. We then slide a 2×2 block across the grid, which results in 5×5 overlapping block positions:

$$\text{HOG features} = \underbrace{(5 \times 5)}_{\text{block positions}} \times \underbrace{(2 \times 2)}_{\text{cells per block}} \times \underbrace{9}_{\text{directions}} = 900$$

Fuzzy classifier protocol. For Task 4, geometric features were extracted directly from original-resolution frontal KDEF images, rather than the FER-compatible 48×48 images. Both acquisition sessions for each identity remained in the same partition. Model selection used training subjects only, preserving the held-out KDEF subjects for final evaluation.

FER2013 class balance. We use the provided split: 28,709 training images, 3,589 validation images (PublicTest) and 3,589 test images (PrivateTest). The classes are strongly imbalanced (Table 5). Happy is the largest class (7,215 training images) while Disgust has only 436 (1.5% of the training set). Task 1 deliberately uses no class weighting, so that the plain baseline can later be compared with balancing strategies in the creative challenge (Task 5).

[Replace: Detected-face failures, deduplication/quality checks, exact train/validation/test protocol, seeds, tools, versions and potential limitations. Provide one dataset example if it supports a claim.]

3 Task 1: FER2013 baseline

Features. Three feature types were extracted, each compared with three classifiers ($3 \times 3 = 9$ models):

- **HOG (900 dimensions):** histogram of oriented gradients on the 48×48 image, 9 orientations, 8×8 pixel cells, 2×2 blocks with L2-Hys normalization. HOG encodes local edge orientation (mouth corners, brows, eye shape, wrinkles) and the block normalization gives some robustness to illumination.
- **Facial landmarks (136 dimensions):** the 68 dlib landmarks, centred and scaled to $[-1, 1]$. The whole 48×48 image is passed to dlib as the face box, so the same procedure can be reused in the real-time demo.
- **PCA (100 dimensions):** raw pixels scaled to $[0, 1]$, projected with whitened PCA. PCA is fitted inside the scikit-learn pipeline on the training data only, which avoids leakage into the validation and test sets.

Classifiers. Multinomial Logistic Regression ($C = 1$, standardized inputs), an RBF-kernel SVM ($C = 1$, $\gamma =$ "scale", standardized inputs) and a Random Forest (300 trees, sqrt features, no scaling). All use a fixed random seed (42).

Results on the validation set. Table 1 shows accuracy, balanced accuracy and macro-F1 for all nine combinations. Macro-F1 was the selection criterion, because it weights all seven classes equally and is therefore not dominated by Happy.

Table 1: FER2013 PublicTest (validation) results for all nine feature–classifier combinations, with default settings. The selected model is in bold. Metric heatmaps are in Appendix A.2.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1
HOG	Logistic Regression	0.435	0.397	0.395
HOG	**SVM**	**0.542**	**0.482**	**0.505**
HOG	Random Forest	0.482	0.414	0.440
Landmarks	Logistic Regression	0.470	0.397	0.394
Landmarks	SVM	0.500	0.409	0.401
Landmarks	Random Forest	0.514	0.456	0.474
PCA	Logistic Regression	0.382	0.299	0.287
PCA	SVM	0.477	0.410	0.430
PCA	Random Forest	0.454	0.395	0.423

Best classifier. The RBF-SVM is the best classifier on two of the three feature types (HOG: 0.505 macro-F1; PCA: 0.430) and the best overall. Its non-linear kernel can model the non-linear relation between local gradient patterns and expression, which the linear Logistic Regression cannot: Logistic Regression is the weakest or near-weakest classifier for every feature type (0.287 to 0.395 macro-F1). The Random Forest is competitive only on landmarks, where it is the best classifier (0.474) and clearly beats the SVM (0.401). A plausible reason, which we have not tested, is that the forest can pick out a few informative coordinates from the 136 landmark values, while the RBF kernel, with standardized inputs and equal weight on all coordinates, is diluted by the uninformative ones. The SVM is also by far the most expensive model to train (438 s on HOG, against 36 s for the forest and 6 s for Logistic Regression).

Best features. HOG is the best representation for Logistic Regression and the SVM (for example 0.505 against 0.430 for PCA and 0.401 for landmarks with the SVM). For the Random Forest, landmarks are the best representation (0.474 against 0.440 for HOG). We attribute the HOG result to its capture of local texture and edge information that distinguishes expressions (open mouth, raised or furrowed brows, wrinkles around the eyes and nose), with some invariance to illumination. Landmarks discard texture, and at 48×48 pixels the dlib detector works with very low resolution, so the landmark positions are noisy. Landmarks are also computed with the full image as the face box, which assumes FER crops are well aligned; this is not always true (see the error analysis below). PCA is a holistic linear projection of raw pixels, so it is sensitive to misalignment, pose and lighting.

Hyperparameter tuning and final test result. Each classifier was tuned on the feature set where it did best on validation. The SVM grid covered $C \in \{1, 5, 10\}$ and $\gamma \in \{0.5, 1, 2\} \times 1/900$, tuned on a 10,000-image subsample of the training set; the Logistic Regression grid covered $C \in \{0.01, 0.1, 1, 10\}$; and the Random Forest grid covered maximum depth (unlimited or 25) and minimum leaf size (1, 3 or 5). The best settings were refit on the full training split and scored on validation: the SVM reached 0.562 macro-F1 ($C = 10$, $\gamma = 1/900$), against 0.505 by default; Logistic Regression ($C = 0.01$) reached 0.411; and the landmark Random Forest (maximum

depth 25, minimum leaf 1) reached 0.477. Each tuned model was then evaluated once on the private test split (Table 2).

Table 2: Final test performance of each tuned classifier on the PrivateTest split, with 95% bootstrap intervals for macro-F1 (1,000 resamples). The confusion matrices are in Appendix A.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1	95% interval
HOG	SVM ($C = 10$)	0.575	0.550	0.568	[0.544, 0.590]
Landmarks	Random Forest	0.517	0.468	0.487	[0.461, 0.512]
HOG	Logistic Regression	0.438	0.376	0.378	[0.355, 0.400]

The test score of the SVM is higher than its validation score (0.568 against 0.562 tuned, and 0.505 default). The paired bootstrap difference between the SVM and the landmark Random Forest is +0.081 with interval $[+0.057, +0.105]$, so the SVM's lead on the test split is unlikely to be noise. The gap between accuracy and balanced accuracy on validation (0.542 against 0.482) is caused mainly by the low recall of Disgust (25%).

Which emotions are hardest to distinguish? Averaging per-class F1 over the nine validation models gives the following ordering from hardest to easiest: Fear (0.26), Disgust (0.27), Angry (0.32), Sad (0.38), Neutral (0.43), Surprise (0.62), Happy (0.64). For the SVM on the test set, F1 is 0.75 for Happy and 0.73 for Surprise, and 0.43 to 0.46 for Angry, Sad and Fear.

- Happy and Surprise are relatively easier to classify. Both have distinctive, large-scale configurations: a raised, widened mouth with visible teeth for Happy, and wide-open eyes and mouth for Surprise. Happy additionally has the most training data.
- Fear, Angry and Sad are harder and are confused with each other and with Neutral. The most frequent errors of the validation SVM are Fear to Sad (24% of all Fear), Sad to Neutral (19%), Angry to Sad (18%), Sad to Happy (17%) and Neutral to Sad (17%); the test set shows the same pattern (Appendix A). These classes are expressed through subtle, low-intensity changes around the brows and eyes, which are hard to resolve at 48×48 pixels. Fear also shares features with Surprise (wide eyes, open mouth), and Sad and Neutral faces are visually close, so Neutral is a frequent "sink" class.
- Disgust is rare (436 training images), so the classifier rarely predicts it. On validation the SVM has precision 1.00 but recall 0.25: when it predicts Disgust it is right, but it misses most Disgust faces. On the test set recall is 0.51, but with only 55 test images this estimate is noisy. Class balancing was tested in Task 5 and did not help.

Evidence from error analysis. Illumination and contrast have a small effect on error. We grouped validation images into quintiles of mean brightness and of contrast (pixel standard deviation) and computed the error rate of the SVM in each group. The error rate is about 0.46 to 0.49 for the three darkest quintiles and about 0.42 to 0.44 for the brightest two, and it rises slightly from about 0.44 to 0.48 as contrast increases (Figure 2). The effect is around five percentage points, so illumination matters but is not the main source of error.

Figure 3 shows recurring problems that are properties of the dataset rather than of the model: *ambiguous or questionable labels* (a screaming infant labelled Angry and predicted Happy; several wide-eyed, open-mouthed faces labelled Neutral or Fear), *occlusion and context* (hands over the face, hats, two faces in the crop), *non-frontal pose and head tilt*, which break the aligned-face assumption of all three feature types, and *population and artefacts* (infants and elderly faces, overlaid text or watermarks). The gallery is a random sample of errors, so it is qualitative evidence, not a count of mislabelled images.

Duplicates. FER2013 contains exact copies of images across splits, and some copies carry different labels, which is direct evidence of label noise. Of the 3,301 non-duplicated test images, the SVM reaches 54.2% accuracy and 0.505 macro-F1; the 288 duplicated test images (8.0%) reach 95.5% accuracy and 0.943 macro-F1, and 96.9% of the duplicates carry the same label as the training copy. The reported test score of 0.568 macro-F1 is therefore optimistic, and the non-duplicated score of 0.505 is the fairer estimate. On validation, 280 of 3,589 images (7.8%) also appear in training, and removing them lowers the SVM's macro-F1 from 0.505 to 0.438. The drop is largest for the Random Forests (0.097 to 0.110), which memorize training samples, and smallest for Logistic Regression and landmark SVM (0.008 to 0.010). We treat cleaning the training set as a data curation step in Task 5.

Summary and limitations. The tuned HOG + RBF-SVM is the best FER2013 baseline (test: 0.575 accuracy, 0.550 balanced accuracy, 0.568 macro-F1; 0.505 macro-F1 without duplicates). HOG is the strongest feature type for Logistic Regression and the SVM, and landmarks are the strongest for the Random Forest; the SVM is the strongest classifier on HOG and PCA. Happy and Surprise are recognized reliably, while Fear, Angry, Sad and Disgust are not. For the demo, prediction time is 0.15 ms per image for Logistic Regression, 32 ms for the SVM, and 27 ms plus 11 ms of feature extraction for the Random Forest. Limitations: (i) the SVM was tuned on a 10,000-image subsample; (ii) the final test was run once per model, with no further selection; (iii) no class

balancing was used, which disadvantages Disgust; (iv) the landmark features depend on a detector applied to 48×48 images with the whole image as the face box.

4 Task 2: KDEF dataset study

In this task we take the best FER2013 representation from Task 1, HOG, retrain the three classifiers on KDEF, and compare the results with FER2013.

The Karolinska Directed Emotional Faces - KDEF was a dataset that was generated in 1998. It has a total of 4900 pictures, of 70 individuals displaying 7 different emotions. The subjects for this dataset was chosen with a specific selection criteria some of which include having no beard, mustache, earrings etc. All subjects were given special gray shirts and all the photos were taken at a distance of 3m from the camera. Each subject was taken photos in 2 sessions, 5 poses for each of the 7 emotions. The pose that is of most interest to us is *straight*.

Unlike the FER2013 dataset (which are 48×48 dimensions), the KDEF dataset has images of dimensions 562×762 and RGB images. To compare the 2 datasets, the KDEF images need to be converted to the FER2013 format: a Haar cascade detects the face, the box is enlarged by 20% so the whole face and some context are kept, and the crop is converted to grayscale, resized to 48×48 and scaled to $[0, 1]$.

The data is split 80/20 by subject i.e out of the 70 subjects, 56 of them are used in the training set, and 14 are in the test set. This is to ensure there is no data leakage into the test set. KDEF is perfectly balanced, with 112 training and 28 test images per emotion (Figure 4.1).

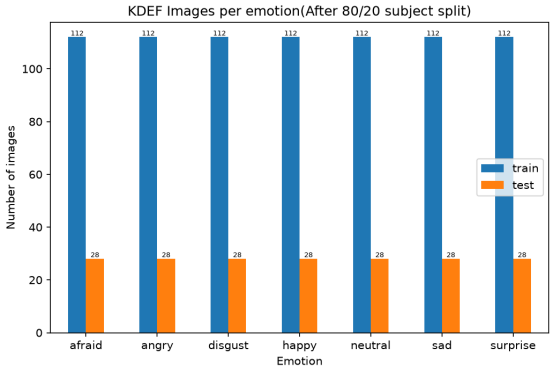


Figure 4.1: Number of KDEF images per emotion in the training (56 subjects) and test (14 subjects) split.

In the previous task, we found out that the best feature representation is Histogram of Oriented Gradients (HOG), so every KDEF image is described with the same 900 HOG features (Section **??**).

4.1 Model training and selection

We train the same three classifiers as in Task 1: Logistic Regression, SVM and Random Forest. For each classifier we run a grid search with 5-fold cross-validation, where the folds are also split by subject (using GroupKFold), so the model is always validated on subjects it has not seen during training.

4.2 Results

Table 4.2.1 shows that all three classifiers score higher on KDEF than on FER2013, even though they are trained on only 784 images instead of about 28,700. The reason is that KDEF is a much cleaner dataset. Every photo is taken in the same studio, with the same lighting and camera position, and the actors show each emotion clearly, so the faces line up well and the expression is almost the only thing that differs between images. The HOG features of two happy faces therefore look very similar, and the classifiers need few examples to learn each emotion. FER2013 images come from the web, with different lighting, head poses, faces partly covered by hands or glasses, and labels that are sometimes wrong, so the classifiers have to learn the same emotion from very different-looking images.

Classifier	Hyperparameter	Values searched
Logistic Regression	Scaler	none, Standard, Robust, MinMax, **PowerTransformer**
	C	**0.01**, 0.1, 1, 10
	20 configurations	
SVM	Scaler	Standard, Robust, **PowerTransformer**
	Kernel	linear, **RBF**, polynomial
	C	0.01, 0.1, 1, **10**
	γ (RBF only)	0.01, 0.001, **0.0001**, auto
	Degree (polynomial only)	2, 3, 4
	96 configurations	
Random Forest	Number of trees	100, **200**
	Maximum depth	3, 5, 8, **10**, 15, 20, unlimited
	Minimum samples per leaf	1, **2**, 5, 10, 20, 50
	84 configurations	

Table 4.1.1: Hyperparameter tuning grid for KDEF: Selected values (highest macro-F1) are in bold

	FER2013		KDEF	
Classifier	Validation	Test	Validation	Test
Logistic Regression	0.395	0.378	0.669	0.581
SVM	**0.505**	**0.568**	**0.679**	0.604
Random Forest	0.440	–	0.611	**0.614**

Table 4.2.1: Macro-F1 with HOG features, for the models trained and evaluated on FER2013 (Task 1) and on KDEF.

4.3 Assignment Questions

Why are KDEF images easier or harder to classify? Overall, KDEF is easier: with the same HOG features, the SVM reaches a macro-F1 of 0.679 in cross-validation and 0.604 on the test subjects. As mentioned before, the images in the KDEF dataset are consistent and is balanced. Each facial attribute is present in a similar pixel vicinity across all images of an emotion. This makes the models learn better, unlike the FER2013 where there is no consistency in the subjects present, alignment and facial features/attributes of the face etc.

To understand how these alignment of these facial features could be of concern for the model, we generated the mean face for each emotion in both datasets. This was done to understand how well positioned each facial feature is in all images across the dataset. The more blurry the images are, the more inconsistent the images are in the dataset for each emotion.

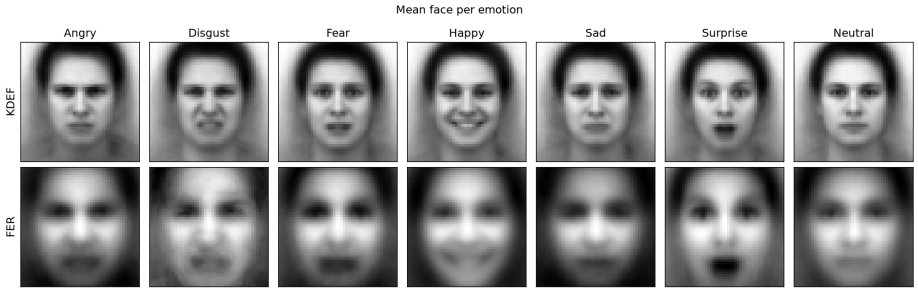


Figure 4.3.1: Mean face per emotion of KDEF and FER2013 training images

Which models change ranking?

On KDEF, Logistic Regression jumps from last to second place in validation, while the Random Forest falls to last. We think this is because the well-aligned KDEF faces make the HOG features almost linearly separable, so a strongly regularized linear model ($C = 0.01$) works well with few training images. Although, the Random Forest performs better in the test set by a small margin, we choose the best model from the validation set.

Which emotions transfer well between datasets? To see which emotions hold up on both datasets, we compare the F1 score per emotion of the KDEF models with that of the FER-trained SVM from Task 1

Rank	FER2013 validation	KDEF cross-validation	KDEF test
1	SVM (0.505)	SVM (0.679)	Random Forest (0.614)
2	Random Forest (0.440)	Logistic Regression (0.669)	SVM (0.604)
3	Logistic Regression (0.395)	Random Forest (0.611)	Logistic Regression (0.581)

Table 4.3.1: FER2013 vs KDEF model comparison

(Table 4.3.2). Happy is the easiest emotion on both datasets (0.93 on KDEF and 0.75 on FER2013 for the SVM), and Surprise also scores well on both (0.62 and 0.73). Both expressions have large, distinctive shapes, a smile or an open mouth with raised eyebrows, which HOG picks up whether the expression is posed in a studio or caught in a web photo. Disgust scores well on both too, but FER2013 contains very few Disgust images (1.5%), so this result is less certain. Fear, Sad and Angry are among the hardest emotions on both datasets, with F1 scores between 0.33 and 0.55.

	KDEF			FER2013
Emotion	Logistic Regression	SVM	Random Forest	SVM
Angry	0.50	0.50	0.65	0.44
Disgust	0.53	0.65	0.69	0.62
Fear	0.41	0.33	0.34	0.46
Happy	**0.88**	**0.93**	**0.77**	**0.75**
Sad	0.51	0.55	0.53	0.43
Surprise	0.63	0.62	0.63	0.73
Neutral	0.61	0.66	0.69	0.55

Table 4.3.2: F1 score per emotion - classifier comparison

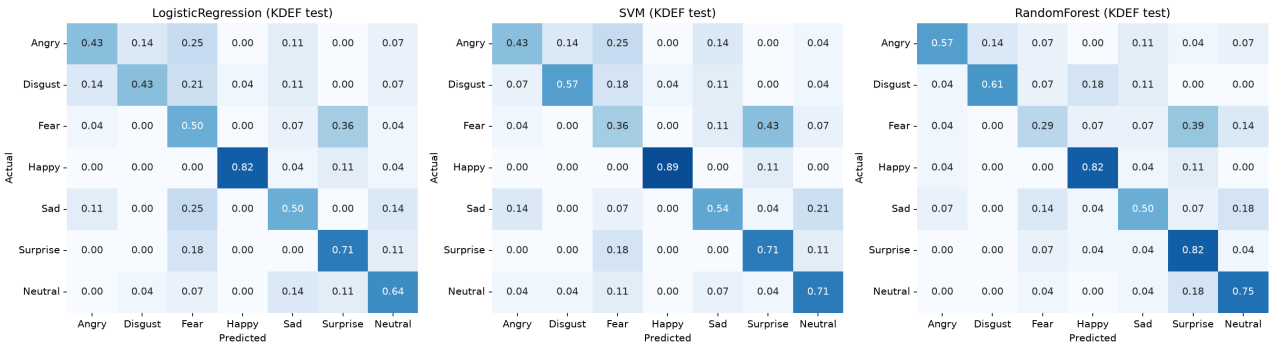


Figure 4.3.2: Confusion matrices(normalized values) for KDEF dataset

The confusion matrices in Figure 4.3.2 show that most mistakes happen between emotions that look alike. Fear is the clearest example: the SVM mistakes more fearful faces for Surprise (43%) than it gets right (36%), and the other two classifiers do the same. This is not surprising, since both expressions have wide-open eyes, raised eyebrows and an open mouth. In the same way, Sad is often mistaken for Neutral (21%), and Angry for Fear (25%) or Disgust (14%). Based on this, we expect Happy and Surprise to transfer best between the datasets, and Fear, Sad and Angry to transfer worst.

5 Task 3: Cross-dataset generalization

In Task 2 our models did well on KDEF, but KDEF is a very clean dataset. Here we test what happens when the model trained on KDEF has to recognise emotions in FER2013, and we look at why it does better or worse.

5.1 Experimental setup

We take the best model from Task 2, the SVM with HOG features, and train it on all 980 frontal KDEF images. We then test it on the full FER2013 dataset (35,887 images). The model has never seen any FER2013 image, so

all of them can be used for testing. Both datasets are preprocessed in the same way and use the same HOG features (Section **??**), and nothing is tuned on FER2013.

To see how big the drop is, we compare the result with two models that are trained and tested on the same dataset: the KDEF model from Task 2 and the FER2013 model from Task 1.

5.2 Results

Trained on	Tested on	Accuracy	Balanced accuracy	Macro-F1
KDEF	KDEF (Task 2, cross-validation)	0.686	0.686	0.679
FER2013	FER2013 (Task 1, PrivateTest)	0.575	0.550	0.568
KDEF	FER2013 (full dataset)	0.225	0.215	0.195

Table 5.2.1: Results of the HOG + SVM model when it is trained and tested on the same dataset, and when it is trained on KDEF and tested on FER2013.

The drop is large (Table 5.2.1). The macro-F1 falls from 0.679 on KDEF to 0.195 on FER2013, and the balanced accuracy of 0.215 is only a little better than guessing at random, which would give about 0.14 (1 in 7). FER2013 itself is not the problem: a model trained on FER2013 reaches 0.568. The problem is that the two datasets are too different.

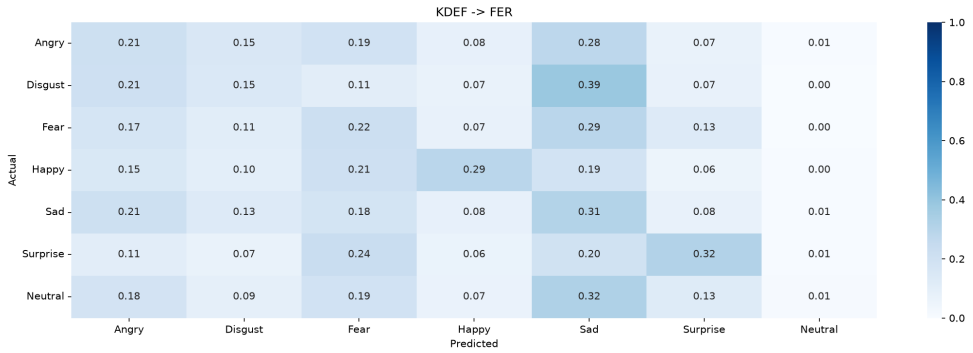


Figure 5.2.1: Confusion matrix of the KDEF-trained SVM on the full FER2013 dataset, normalised per row.

The confusion matrix (Figure 5.2.1) shows what goes wrong. The model puts most faces into just a few emotions, mainly Sad, Fear and Angry. For example, 39% of the disgusted faces and 32% of the neutral faces are predicted as Sad. Neutral is almost never predicted at all, even though it is one of the most common emotions in FER2013. Only Happy and Surprise are recognised a bit more often than the rest.

5.3 Assignment Questions

What adaptations did you have to make to run the model? The two datasets have different formats, so we first made them the same. KDEF photos are large colour images, so we cut out the face with a 20% margin, turned it into grayscale, resized it to 48x48 pixels and scaled the values to between 0 and 1. FER2013 images already have this size, so they only needed rescaling. Both datasets then go through the same HOG extraction. KDEF names its emotions with words and FER2013 with numbers, so we matched them up (for example, "afraid" in KDEF becomes "Fear"). All seven emotions exist in both datasets. We did not tune anything on FER2013, so the test stays fair.

What is domain shift? Domain shift means that the data a model is tested on is different from the data it was trained on. The model learns patterns from its training data, but those patterns may not hold for the new data. KDEF and FER2013 differ in three ways:

- **How often each emotion appears.** KDEF has the same number of images for every emotion. In FER2013, 25% of the images are Happy and only 1.5% are Disgust (Figure 5.3.1, left).
- **What the images look like.** KDEF photos are all taken in the same studio, while FER2013 images come from the web. KDEF images are darker, have less contrast and look much more alike (Figure 5.3.1, right). The mean faces in Figure 4.3.1 also show that KDEF faces are always in the same place in the image, while FER2013 faces are not.
- **How the emotions are shown.** KDEF actors pose each emotion clearly, while FER2013 faces often show subtle, natural expressions, and some of them are labelled wrongly.

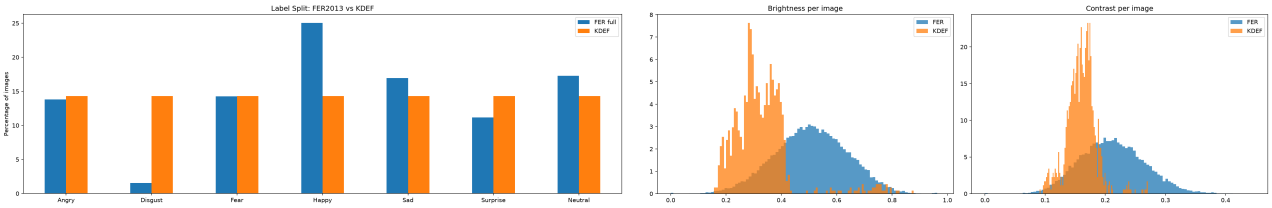


Figure 5.3.1: Left: share of each emotion in FER2013 and KDEF. Right: brightness and contrast of the images in both datasets.

To see whether the model also notices these differences, we trained a simple classifier to guess whether an image comes from KDEF or FER2013, using only its HOG features. With 980 images from each dataset, it guesses right 98.8% of the time (balanced accuracy). So, in the features our model uses, the two datasets look completely different.

Why does performance degrade? The model only learned from 980 studio photos of 70 people looking straight into the camera. FER2013 has many things it has never seen: turned heads, hands in front of the face, children, tilted or blurry photos and all kinds of lighting (Figure 5.3.2). HOG also looks at fixed parts of the image, so when the eyes and mouth are in a different place, as in many FER2013 images, the features no longer match what the model learned. Overall brightness matters less, because HOG normalises it away, but shadows and noise still add edges that have nothing to do with the expression. Finally, the model does not know that some emotions, like Happy, are much more common in FER2013 than others.

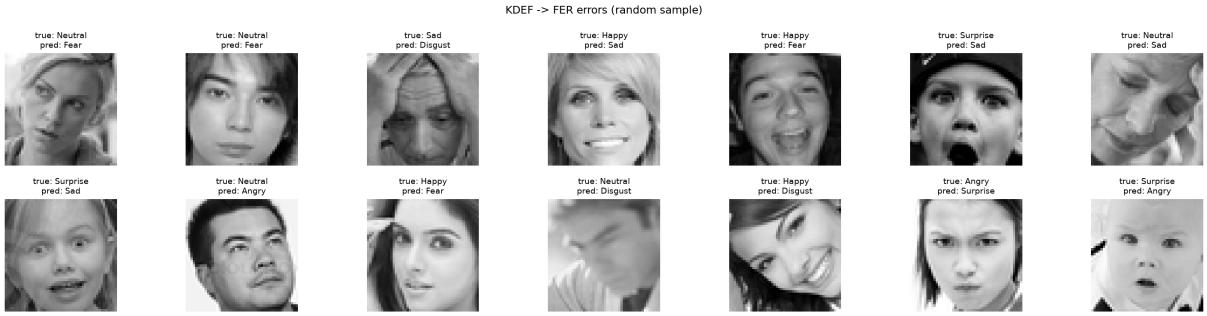


Figure 5.3.2: Random FER2013 images that the KDEF-trained model gets wrong, with the true and predicted emotion.

	Happy	Surprise	Sad	Angry	Fear	Disgust	Neutral
F1 on KDEF (cross-validation)	0.89	0.71	0.50	0.66	0.46	0.69	0.66
F1 on FER2013	0.38	0.31	0.24	0.19	0.18	0.04	0.02

Table 5.3.1: F1 per emotion of the KDEF-trained SVM on KDEF and on FER2013.

Some emotions suffer more than others (Table 5.3.1). As we expected in Task 2, Happy and Surprise hold up best, because a smile or an open mouth looks much the same in a studio photo and in a web photo. Disgust and Neutral do well on KDEF but fail almost completely on FER2013. A real neutral face, with natural shadows and a relaxed mouth, apparently looks sad or scared to a model that has only seen KDEF's very clean neutral faces.

Which features transfer best? We only used HOG features, so we checked which of the 900 HOG features transfer best. For each feature, we computed its average value for every emotion, once in KDEF and once in FER2013, and measured how similar these two patterns are with a correlation. A feature with a correlation close to 1 goes up and down with the emotions in the same way in both datasets, so what the model learned from it still works on FER2013.

Figure 5.3.3 shows that the features around the eyes, mouth and cheeks transfer best, with correlations up to 0.95. These are exactly the parts of the face that show emotions. The features at the edges of the image, which cover hair, background and the outline of the face, transfer worst, and some even behave the opposite way (as low as -0.88). This matches the alignment problem: the centre of the face is in roughly the same place in both datasets, but the edges are not. We did not test other feature types here. We would expect HOG to transfer better than PCA on raw pixels, which depends directly on the lighting of the KDEF photos, while facial landmarks ignore lighting but are hard to find reliably in a small 48x48 image.

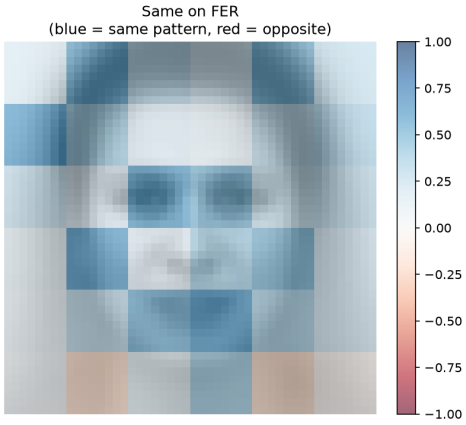


Figure 5.3.3: How well the HOG features in each part of the face transfer from KDEF to FER2013 (blue: same pattern in both datasets, red: opposite pattern).

6 Task 4: Explainable fuzzy classifier

Features and protocol. We extracted seven geometric features from the 68 dlib facial landmarks: mouth width, mouth opening, left and right eye opening, eyebrow height, mouth width-to-height ratio (*smile ratio*), and mouth-corner asymmetry. Most distances were normalized by face width, while eye opening was normalized by eye width. All 980 frontal KDEF images were processed successfully, without landmark detection failures.

The predefined subject-independent split contained 56 training subjects (784 images) and 14 held-out test subjects (196 images). Within the training partition, 44 subjects (616 images) were used for fitting and 12 (168 images) for validation. MinMax scaling was fitted only on the corresponding training partition.

Fuzzy model. We trained seven one-versus-rest Takagi-Sugeno fuzzy models with PyFUME, one for each emotion. The standard settings were modified, which changes PyFUME's default rule simplification and holdout. The relevant parameters altered are: $setnes_threshold = 1.0$, $merge_threshold = 1.0$ and $data_split_method$ = "*no_split*".

Each model uses Gaussian membership functions in its rule antecedents and a learned linear consequent. The numerical outputs are combined through fuzzy inference, and the emotion with the highest output score is selected. These scores are not calibrated probabilities.

We compared two and three clusters per emotion using macro-F1 on validation subjects. The three-cluster configuration was selected before testing and subsequently refitted on all 784 training images.

Table 3: Subject-independent KDEF fuzzy classification results.

Split	Clusters	Images	Acc. (%)	Bal. acc. (%)	Macro-F1
Validation	2	168	71.43	71.43	0.6881
Validation	3	168	73.21	73.21	0.7087
Final test	3	196	63.27	63.27	0.6082

Rule interpretation. The final system contains 21 fuzzy rules, three per emotion. A representative rule from the happiness model is:

> *Happy, Rule 1 (abridged):* IF mouth width belongs to cluster 1 (Gaussian centre 0.762) AND mouth opening belongs to cluster 1 (centre 0.204), together with the remaining geometric conditions, THEN calculate the happiness score using the fitted linear consequent.

This rule dominated 94.6% of happy training images but only 2.2% of other images. Its relatively wide-mouth configuration is consistent with smiling. The strongest rules for fear and sadness showed weaker separation, and overlapping membership functions limit their linguistic interpretation.

Evaluation and limitations. On unseen subjects, happiness (28/28), surprise (26/28) and disgust (23/28) achieved the highest recall. Fear (7/28) and sadness (8/28) were more difficult: 14 fearful faces were classified as surprise, and 12 sad faces as neutral. The roughly ten-percentage-point accuracy decrease from validation to testing indicates limited generalization across individuals. Although the fuzzy antecedents are inspectable, the linear consequents and interactions between the seven models complicate explanations of individual predictions.

For comparison, Table 4.2.1 reports the Task 2 classifiers trained on HOG features and evaluated on the same 14 held-out KDEF subjects (196 images). The fuzzy model reaches a macro-F1 of 0.608, slightly above the SVM selected on validation (0.604) and below the Random Forest (0.614), which scored best on the test subjects but was not selected on validation. Its accuracy of 63.27% is the highest of the three (SVM 60.20%, Random Forest 62.24%, Logistic Regression 57.65%). These gaps correspond to a few images out of 196, so the fuzzy model performs comparably while using seven interpretable measurements instead of 900 HOG features. Its drop from validation (0.709) to test (0.608) is larger than the SVM's (0.679 to 0.604), which is consistent with the limited generalization noted above.

7 Task 5: Creative challenge

We started from the Task 1 baseline and tested one change at a time, using a fast Logistic Regression and the validation split (PublicTest) to decide. The test split was used once, for the final model.

Changes tested.

- **Feature engineering (kept).** The 136 landmark values were appended to the 900 HOG features (1,036 features). HOG describes texture around the mouth and eyes, and landmarks describe the shape of the face, so the two carry different information.
- **Data cleaning (kept).** Training images were fingerprinted with MD5 hashes. Duplicates with the same label were reduced to one copy, and images whose copies disagreed in label were removed (107 images), reducing the training set from 28,709 to 27,434 images. The gain is within the noise level (Section 3).
- **Class balancing (rejected).** Balanced class weights lowered both macro-F1 and Disgust F1, so the switch was left off.
- **Augmentation (not run).** Horizontal-flip copies were considered but not run. They would double the training set, and the RBF-SVM training time is already the limiting factor.

Table 4: Validation (PublicTest) ablation with Logistic Regression. Each row adds one change to the rows above it.

Variant	Accuracy	Macro-F1	Disgust F1	Training images
Baseline (HOG)	0.435	0.395	0.324	28,709
+ landmarks	0.497	0.453	0.325	28,709
+ cleaning (kept)	0.500	0.455	0.325	27,434
+ class balancing (rejected)	0.469	0.424	0.242	27,434

Final model. The tuned Task 1 SVM ($C = 10$) was retrained on the kept changes: 1,036 features and the cleaned training set. Because the feature count rose from 900 to 1,036, γ was rescaled by 900/1036 to keep the kernel width comparable ($\gamma \approx 9.7 \times 10^{-4}$). This rescaling is a heuristic and was not tuned. On validation, the model reaches 0.607 accuracy and 0.598 macro-F1, against 0.562 for the tuned Task 1 SVM. On the test split, evaluated once, it reaches 0.594 accuracy, 0.572 balanced accuracy and 0.592 macro-F1, against 0.568 for the Task 1 SVM.

Interpretation. Adding landmarks is the main gain (+0.058 macro-F1 for Logistic Regression), and it is consistent with the landmark-based Random Forest in Task 1, which is the best classifier on landmarks. Cleaning changes the score by 0.002, which we treat as no effect. Class balancing hurt, even though Disgust recall was low (0.25) in Task 1. The notebook does not analyze why, so we leave the mechanism open. The test gain of 0.024 macro-F1 over Task 1 is not tested for significance, and the Task 1 bootstrap interval for macro-F1 is about ±0.023. Two limitations apply. The ablation uses Logistic Regression, not the final SVM, and each step is a single run. The test split still contains the 8% of images that also appear in training (Section 3). KDEF-derived features were not used in the final model.

8 Task 6: Real-time webcam demo

Our demo adapts the best selected model into a command-line executable Python file that opens the laptop webcam and performs live emotion recognition. We have chosen the model from **Task 4 - Fuzzy Classifier**, which was trained based on the *KDEF* database.

We added a **README** file in the root folder of the submitted files explaining how to run the demo in the terminal. The same results can be obtained by running the notebook `notebooks/06_task6_webcam_demo.ipynb`.

The seven features used in the live detection are identical to the ones reported in section 6. Detection, however, was done differently, since training upsampled once on the original *KDEF* photos. The demo uses no upsampling on the downscaled frame.

We performed an empirical evaluation where the following protocol was followed:

- Refer to an emotion from a **KDEF** reference
- Mimic the emotion and start the simulation
- Run the simulation for a while without changing the emotion displayed
- Compute the results

The final experiment data can be visualised in 7 in appendix A. The experiment summaries show that happy, sad, surprise, neutral, disgust and angry were classified in $> 80\%$ of the cases. Fear, on the other hand, was extremely difficult to trigger, and it was confused mostly with neutral and sad. Obviously, this experiment does not constitute a reliable evaluation metric for the live demo classifier. For that, it would be necessary to expose the model to data with emotions labelled properly.

Challenges and limitations: To improve throughput, our code downscales the whole frame to 640 px in width (for detection only). Landmarks are then taken from the full-resolution frame. This modification substantially increased the processing fram rate from 1.17 ($693.03ms$) to 13.71 processed FPS ($17.28ms$ average detection), running on MacOS.

In terms of luminosity, camera permissions and detection robustness, we did not experience any major issues, but mostly of the development for that section was done in MacOS. We foresee that other operating systems might add extra challenges during execution, which is a limitation the team could not address in this particular assignment.

9 Discussion, limitations and conclusion

[Replace: Synthesize: best features/classifiers on each dataset; relative difficulty of each emotion; changes of ranking; strongest and weakest transfers; fuzzy interpretability versus performance; benefits and costs of creative improvements; webcam readiness. Contrast dataset-specific and common limitations. Draw conclusions only from actual measurements and concrete error examples.]

Declaration of AI tool use

Technology Statement: During the preparation of this work, we used OpenAI ChatGPT (GPT-5.6 Sol), Google Gemini (3.1 Pro) and Claude Sonnet 5 Medium in order to brainstorm assignment direction, support on coding and discuss specific technical concepts to support the team along the development. The following parts of the assignment were affected/generated by AI tool usage: TASK 1 / TASK 2 / TASK 3 / TASK 4 / TASK 5 / TASK 6. After using this tool/service, Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto take full responsibility for the content of their work.

References

Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska Directed Emotional Faces: A validation study. *Cognition & Emotion*, *22*(6), 1094–1118. https://doi.org/10.1080/02699930701626582

Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In M. Lee, A. Hirose, Z.-G. Hou, R. M. Kil, D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, & G. Weikum (Eds.), *Neural Information Processing* (pp. 117–124, Vol. 8228). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-42051-1_16

A Confusion matrices for reported models

Confusion matrices for every model reported in the main text, exported by the notebooks. FER→KDEF has not been run yet: *[Replace: add its confusion matrix once run]*.

A.1 Dataset and error-analysis figures (Tasks 1 and 3)

Table 5: FER2013 class counts per split.

Split	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Train	3,995	436	4,097	7,215	4,830	3,171	4,965
Validation	467	56	496	895	653	415	607

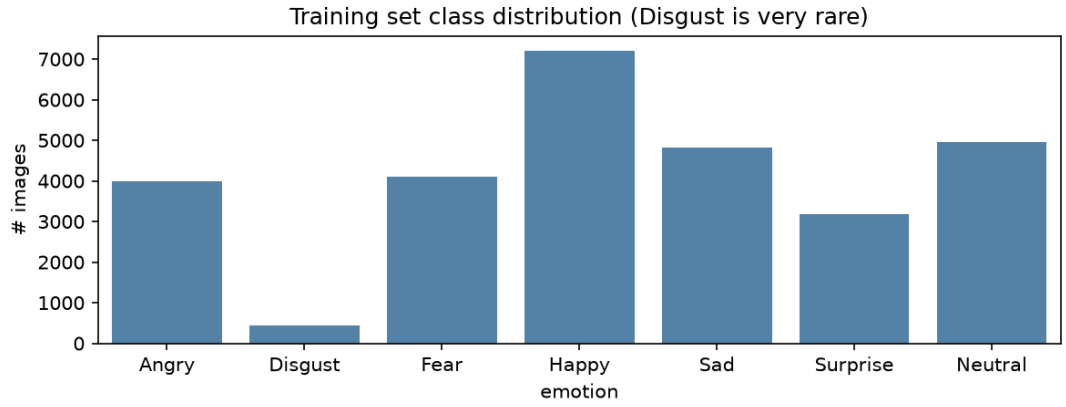


Figure 1: FER2013 training set: class distribution.

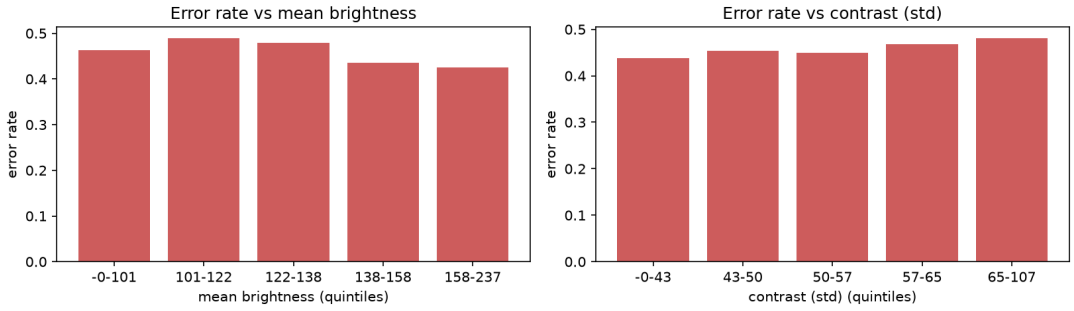


Figure 2: Error rate against lighting (HOG + SVM, validation split).

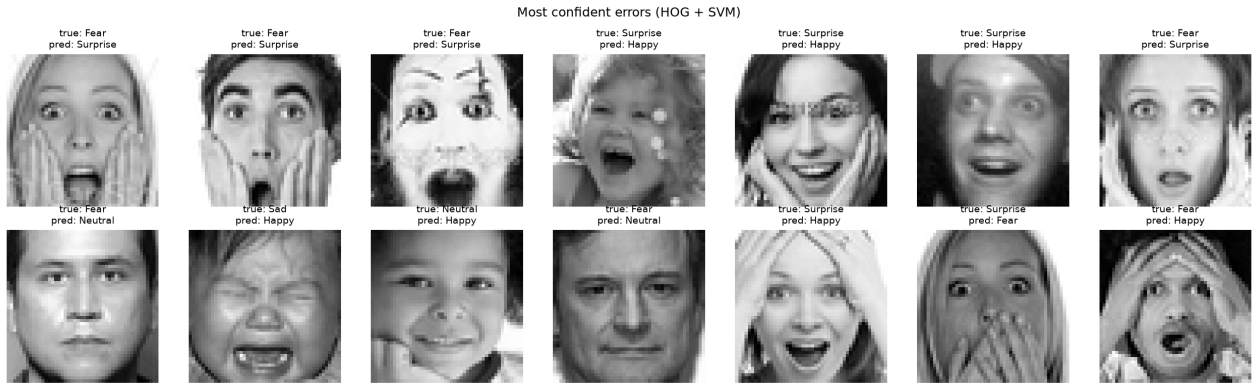


Figure 3: Random sample of misclassified FER2013 validation images.

A.2 Validation metric heatmaps (FER2013)

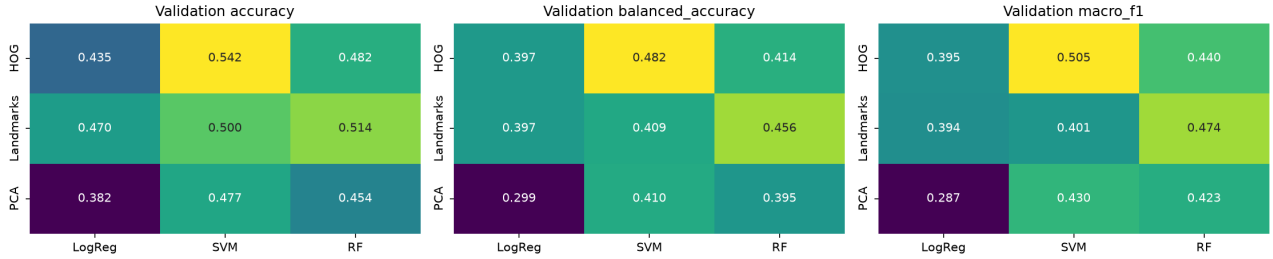


Figure 4: Metric heatmaps of each feature–classifier combination on the FER2013 PublicTest (validation) split.

A.3 FER2013 baseline models (PublicTest; tuned models on PrivateTest)

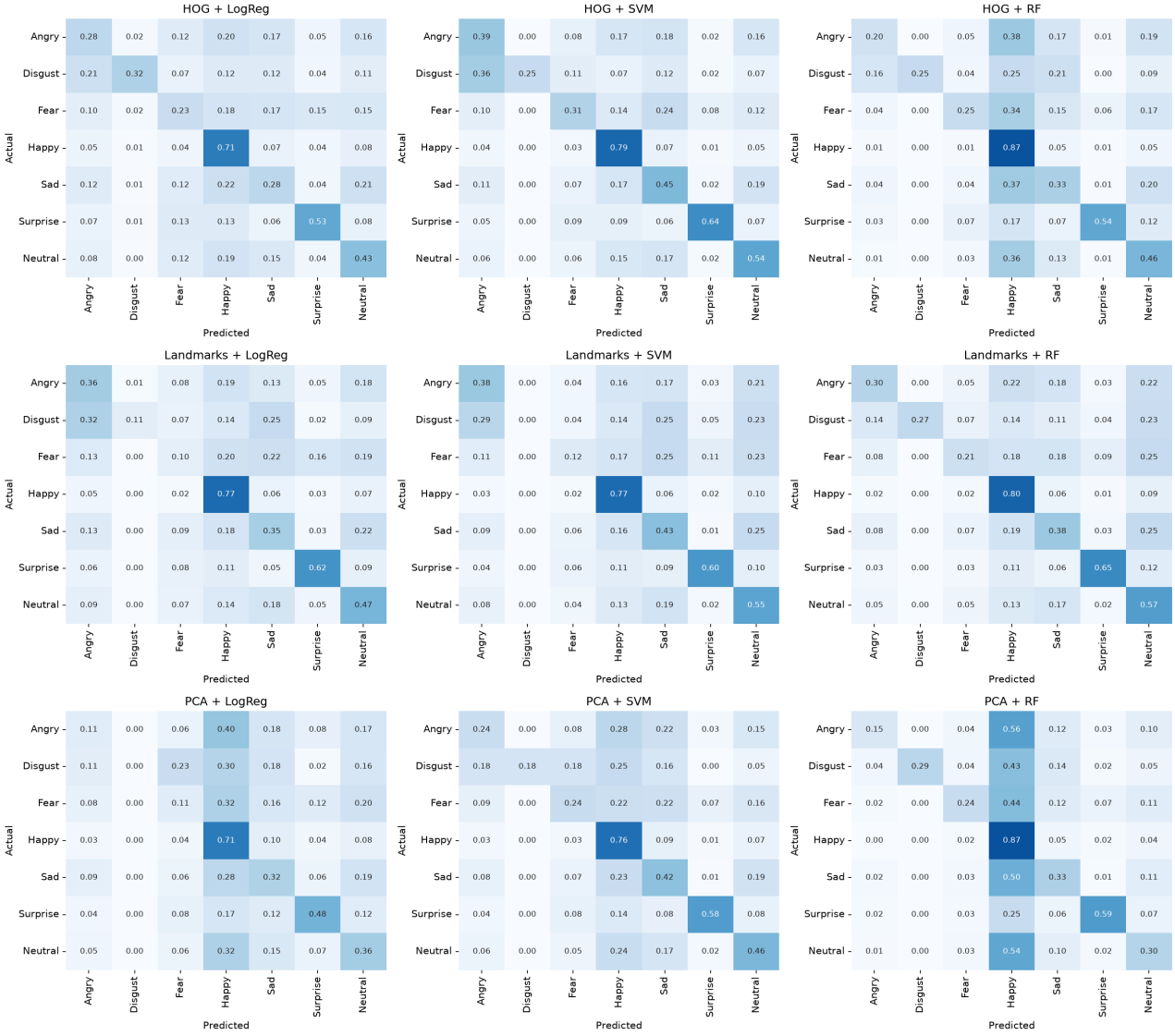


Figure 5: Confusion matrices of all nine FER2013 baseline models on the PublicTest (validation) split. Rows are true labels and columns are predictions; values are row-normalized.

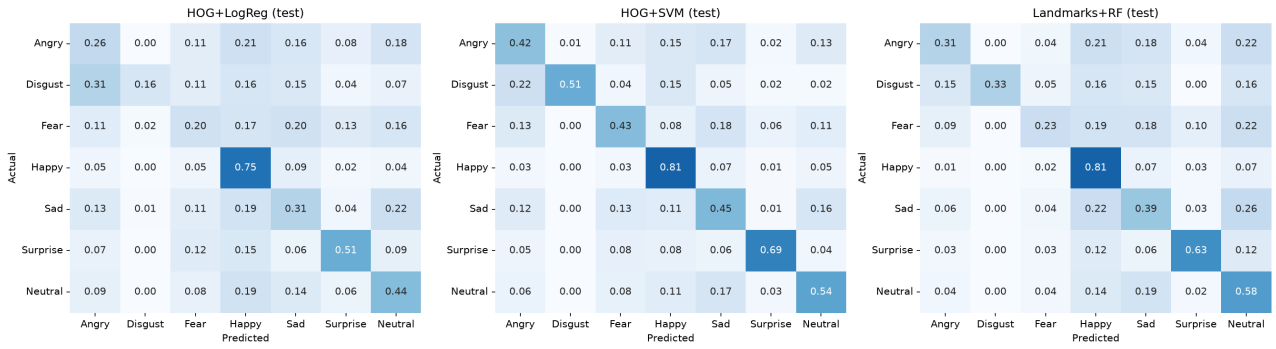


Figure 6: Confusion matrices of the three selected FER2013 models on the PrivateTest split, used for the final evaluation (Table 2). Rows are true labels and columns are predictions; values are row-normalized.

A.4 KDEF study, cross-dataset and creative models

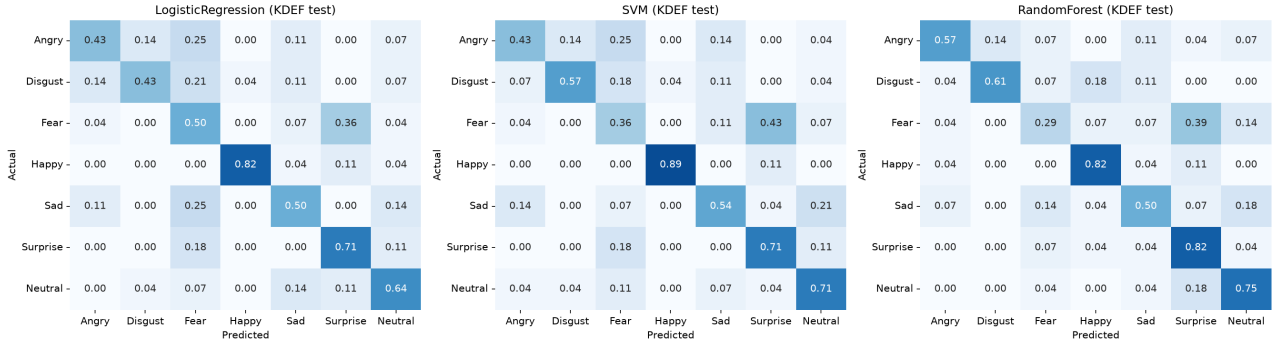


Figure 7: KDEF test subjects (196 images): Logistic Regression, SVM and Random Forest with HOG features, left to right.

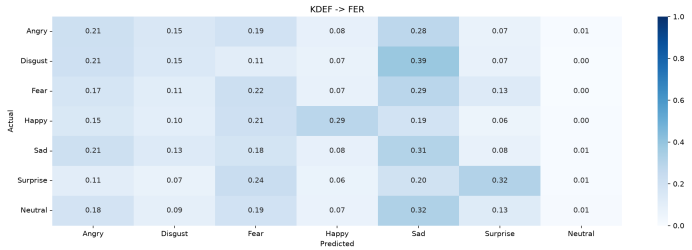


Figure 8: KDEF→FER: the Task 2 SVM trained on all KDEF frontal images, evaluated on all FER2013 images. Rows are true labels, row-normalized.

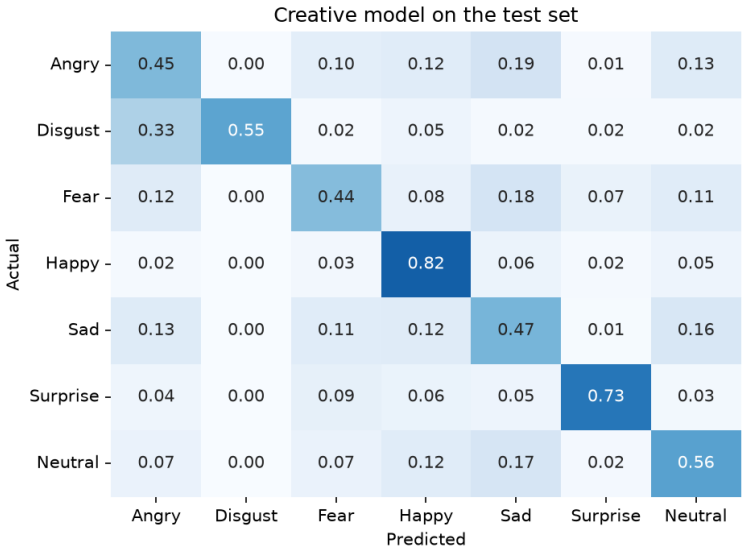


Figure 9: Creative-challenge final model (SVM on HOG + landmarks, cleaned training set), PrivateTest split.

A.5 Fuzzy classifier: held-out KDEF subjects

Table 6: Final fuzzy classifier confusion matrix (counts; 196 test images from 14 unseen subjects). Rows are true labels and columns are predictions.

True / Pred.	Af	An	Di	Ha	Ne	Sa	Su
Afraid	7	1	2	0	2	2	14
Angry	0	14	10	0	2	2	0
Disgust	0	3	23	1	0	1	0
Happy	0	0	0	28	0	0	0
Neutral	0	3	0	0	18	6	1
Sad	0	6	2	0	12	8	0
Surprise	0	0	0	0	1	1	26

Af: afraid; An: angry; Di: disgust; Ha: happy; Ne: neutral; Sa: sad; Su: surprise.

A.6 Empirical Analysis - Live demo detection using Fuzzy model.

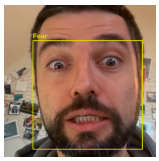
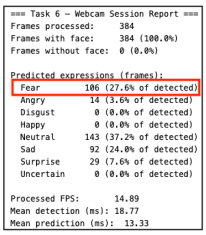
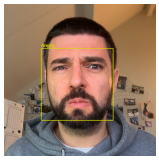
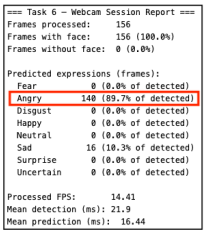
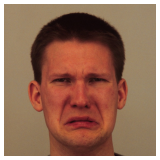
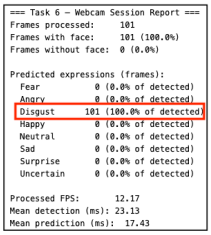
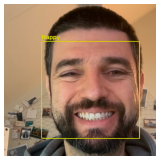
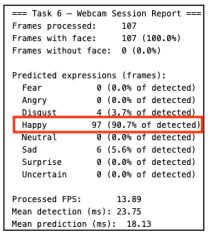
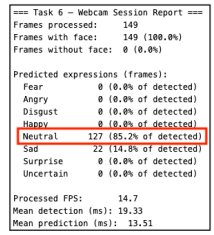
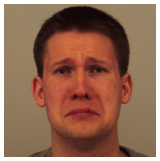
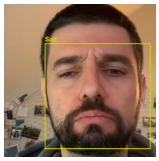
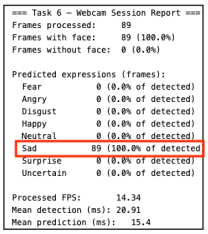
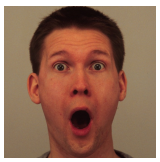
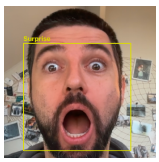
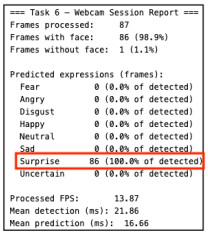
Posed expression	KDEF reference	Webcam capture	Prediction
Fear			
Angry			
Disgust			
Happy			
Neutral			
Sad			
Surprise			

Table 7: Qualitative comparison of posed KDEF expressions, corresponding webcam expressions, and fuzzy classifier predictions. KDEF subject M11, acquisition A, frontal pose. Webcam participant was the group member Henrique da Silva Couto. This illustrative experiment does not establish classification accuracy.